Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

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Abstract

Bayesian networks (BNs) can support explainable software engineering decisions, but constructing their conditional probability tables (CPTs) is costly. This study explores the use of large language models (LLMs) to support constructing CPTs through direct elicitation and using a popular semi-automatic CPT construction technique, the Ranked Nodes Method (RNM). We evaluate 61 ordinal conditional nodes from seven software engineering BNs, crossing elicitation method, structured prompting, and qualitative retrieval-augmented generation (RAG), with five replications requested from GPT-6 Astra at medium effort. Reference CPTs remain hidden during elicitation and experimental fitting. The primary comparison uses the 25 multi-parent nodes with parent configurations not elicited by RNM: 14 from a process-management BN and 11 from two code-review BNs. RNM lowers total variation from 0.3184 to 0.2761 for the RNM-derived process reference, but raises it from 0.2223 to 0.2517 for the code-review references. Both contrasts survive Holm-adjusted paired t-tests; neither rejects an exact sign test with five replications. Across complete-table evaluations, structured prompting and qualitative RAG provide no consistent improvement. RNM requests 40.1% fewer distributions and uses 12.1% fewer output tokens, but only 0.61% fewer total tokens. These exploratory findings support evaluating compact elicitation together with its reconstruction assumptions: fewer requested judgments do not guarantee closer reference agreement or proportional computational savings.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Software engineering (SE) decisions about release readiness, quality, and process improvement require more than a prediction: decision makers need to understand the evidence and assumptions behind a recommendation. Transparency and explainability help them assess whether a model is appropriate for the decision and context. A review of 108 studies documents this concern across 23 SE tasks [3]. Data availability presents another obstacle. Training machine learning and deep learning models requires relevant observations, which may be scarce for managerial decisions. Rique et al.'s practitioner study identifies data unavailability among the challenges of adopting software analytics and notes that collecting the required data can be infeasible [22].

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Bayesian networks (BNs) address these needs by explicitly representing dependencies and uncertainty, allowing expert knowledge to complement available observations. Their graphs and probabilities expose assumptions that stakeholders can inspect and revise. In SE, BNs have supported early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [20]. A systematic mapping of 106 studies identifies software management and quality as prominent applications [23]. Their usefulness nevertheless depends on credible structure and parameters.

Constructing conditional probability tables (CPTs) is a substantial part of this modeling effort: the number of conditional distributions grows multiplicatively with the states of each additional parent [11]. Semi-automatic techniques reduce how much experts must specify. Examples include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [19], probability generation through Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. These approaches reconstruct probabilities from a smaller set of judgments or parameters. They reduce elicitation burden, but still require knowledgeable input, and their simplifying assumptions can affect fidelity [2].

Large language models (LLMs) are now being explored as a source of probabilistic knowledge for BNs. Nafar et al. directly elicit numerical parameters [18]; Gottal and Matthes reconstruct CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. use retrieval-augmented generation (RAG) to support expert BN modeling [6]. These studies motivate a further question: should an LLM specify every CPT row, or provide selected judgments to a semi-automatic construction method? Reduced elicitation could save computation, yet reconstruction could introduce error even when the elicited judgments are useful.

We investigate this tradeoff for RNM, which represents ordinal relationships using weighted parent functions and truncated-normal distributions [11, 15]. Our pipeline elicits distributions for selected parent configurations, called anchors, and an aggregation family, then fits the remaining parameters. We compare it with direct CPT elicitation on 61 ordinal conditional nodes from seven SE BNs. Eight conditions cross method, baseline or structured prompting, and absence or presence of qualitative RAG, with five replications requested from GPT-6 Astra at medium effort. Reference CPTs are hidden during elicitation and experimental fitting. We ask:

- **RQ1—Reference agreement:** Does RNM reduce total variation distance relative to direct elicitation on the same non-anchor configurations?
- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?
- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

The primary comparison evaluates the direct and RNM pipelines on identical non-anchor configurations of 25 eligible multi-parent nodes; complete-table and effort analyses cover all 61. We contribute a controlled comparison of reference agreement and elicitation effort, with offline diagnostics separating errors in elicited judgments from changes introduced by reconstruction. RNM's benefit varies across references, and fewer requested distributions yield limited total-token savings. The SE restriction and statistical analyses are post hoc, so the findings concern repeated executions on these references. Explainability motivates the application; practitioner and downstream benefits remain open.

2 Related Work

LLM-assisted BN construction varies in what the model supplies, what evidence it receives, and how the resulting probabilities are validated. Table 1 compares the closest CPT-parameterization studies and two adjacent approaches to BN modeling and inference. These distinctions matter because agreement with a reference CPT, usefulness to a modeler, and downstream predictive performance answer different questions.

2.1 Direct elicitation and evidence-based reconstruction

Nafar et al. provide the closest broad benchmark of numerical LLM parameterization [18]. Their statewise strategy lowers median BN KL divergence for seven of eight models while requiring more queries. Their FullDist baseline requests one conditional row per query; our direct baseline requests a complete CPT in one call. Their results therefore motivate studying elicitation granularity, but do not resolve whether omitting entire parent configurations is beneficial.

Gottal and Matthes instead reconstruct CPTs from numerical summaries, comparing algorithms with direct LLM output [12]. Their empirical constraints differ from our qualitative RAG packets, which exclude reference probabilities and numerical parameters. In a recent preprint, Rahul and Chowdhury aggregate responses from a virtual persona panel [21]. Their consistency checks include refitting data sampled from the elicited BN itself; these checks cannot establish agreement with an independent reference. The study nevertheless reports computational effort, showing why probability generation should be assessed alongside its resource demands.

2.2 Modeling support and probabilistic reasoning

BIRD illustrates a neurosymbolic approach [7]: LLM judgments are converted into parameters constrained by a BN representation [9]. It is a relevant predecessor for combining elicitation and fitting, although relative support judgments and downstream accuracy do not establish exact CPT recovery. Cypko et al.'s clinical RAG study similarly supports the feasibility of assisting BN modelers without validating complete numerical tables [6]. AutoElicit elicits priors for predictive-model parameters and evaluates prediction after Bayesian updating [4]; its endpoint is distinct from reference-CPT agreement. We evaluate that narrower endpoint with fixed graphs and definitions, without inferring explanation quality or practitioner benefit.

2.3 Partial elicitation and the remaining comparison

RNM constructs ordinal CPTs from weighted parent functions and truncated-normal distributions [11, 15]. InterBeta offers another route to reconstructing tables from partial expert input [1]. Blomaard et al. compare RNM, functional interpolation, and InterBeta using known CPTs to optimize parameters [2]; a recent preprint compares human-elicited WSA and RNM pipelines in software decisions [19]. Our experimental RNM fit instead uses only LLM anchors. Reference-informed fits are separate approximation diagnostics.

These strands leave a specific question: can sparse LLM elicitation retain reference agreement while reducing effort when a constrained method supplies the remaining CPT rows? We examine it through matched direct/RNM comparisons, prompt and qualitative-RAG treatments, and measured token usage. Separating errors in the original anchors from changes introduced by fitting helps distinguish inadequate LLM judgments from reconstruction mismatch. This design tests an accuracy–effort tradeoff on SE references; it does not assume that an ordinal CPT is well represented by RNM or that fewer requested distributions imply proportional token savings.

3 Method

We conducted a repeated, factorial comparison of direct CPT elicitation and LLM-assisted RNM reconstruction. The graph, variable meanings, state mappings, elicitation templates, and numerical implementation were fixed before formal collection. Our hypotheses predicted lower non-anchor error with RNM (RQ1), improved agreement from structured prompting and qualitative evidence (RQ2), and fewer requested distributions and output tokens with RNM (RQ3). The subsequent restriction to SE and the statistical test plan were developed after inspecting the broader results. We therefore report an exploratory analysis of a fixed benchmark, rather than a preregistered confirmatory experiment.

3.1 Sample and reference construction

The source pool was an availability-based collection of accessible published models and an existing Scrum model, screened for usable CPTs and definitions; it was not a systematic or representative sample of SE applications. Table 2 describes the 61 retained conditional targets in seven BNs from five model/source families. Eligibility required a complete numerical reference, interpretable definitions, and an ordinal interpretation for the child and every parent. Binary variables qualified only when their meanings supported an order; nominal families and Boolean gate chains were outside scope. The references cover Scrum practices, Chrome code review and defects [14], ERP data quality for advanced planning and scheduling (APS) [13], software maintenance logs [8], and software strategic indicators (SESSI) [16]. The two Chrome versions share one publication family, as do the two software-log BNs.

The Scrum reference is the full model: 25 conditional targets and 17 fixed roots, extending beyond the teamwork fragment used during development. The accompanying thesis descriptions and company questionnaire supplied construct meanings; English endpoint paraphrases operationalized them for this experiment, without

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM pipeline omits selected rows and reconstructs them.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [21] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; 25 primary targets; reference distances, five replications, tokens, and fitting time.	Compares both pipelines on identical non-anchor rows; separately compares raw and fitted judgments on identical anchor rows.

Table 2: Retained targets and requested rows per prompt/evidence combination and replication. Primary targets have at least two parents and at least one non-anchor row.

Network	Targets	Primary	CPT rows	Anchors
Scrum	25	14	249	123
Chrome 40	15	6	78	57
Chrome 42	12	5	63	46
APS	3	0	8	8
SESSI	1	0	3	3
Software logs 3	4	0	16	12
Software logs 4	1	0	4	3
Total	61	25	421	252

claiming a newly validated generic instrument. The XML supplies the numerical reference and stores functions, weights, variances, and expressions rather than complete CPTs. We reconstructed its 249 conditional rows using equal-width ordinal intervals and five endpoint-inclusive integration points per parent state. The reconstruction explicitly preserves two piecewise expressions and the reversed storage order of a volatility variable. Numerical checks and agreement with the earlier fragment support implementation consistency, but do not certify equivalence to a full AgenaRisk CPT export. This RNM-derived reference is reported separately because its construction favors compatibility with an RNM representation, and the development work already involved the Scrum model.

External references use complete exported tables with checked dimensions, finite nonnegative entries, normalization, and parent/child indexing. Source papers informed definitions and semantic ordering; the frozen exports determine the numerical reference. The SESSI target uses the complete originally elicited table in Appendix A3, rather than claiming recovery of the later recalibrated model. Six other Mendes BNs were excluded because complete CPTs were unavailable. The original collection contained 77 targets; the disclosed SE restriction removes 14 marine-risk and two digital-twin targets and drops the domain-comparison RQ. Existing responses were retained without recollection. Earlier pilot and smoke-test answers are excluded. The largest retained CPT has 27 rows.

The primary subset contains 14 Scrum and 11 Chrome targets. The remaining 36 targets support complete-table and effort analyses, but do not satisfy the multi-parent compression criterion. References are contextual model parameters, not established probabilities for arbitrary companies.

3.2 Elicitation conditions and qualitative evidence

The $2 \times 2 \times 2$ design crosses method (direct/RNM), prompt (baseline/structured), and evidence (no RAG/qualitative RAG). All conditions receive the same relevant expert-role description, setting, child and parent definitions, state labels, storage order, semantic

ranks, and parent influence signs. The requested distribution conditions on *all* listed parent states and averages over unspecified circumstances. It neither fixes omitted circumstances to a middle state nor assumes independence. A persona supplies a professional perspective, without asserting access to company observations.

The baseline asks for normalized distributions in a fixed JSON schema. The structured bundle additionally makes conditioning explicit, frames probabilities as expected counts among 100 hypothetical comparable cases (fractional counts allowed), distinguishes evidence from assumptions, and requests consistency and normalization checks. These cases are a judgment aid, not observations. A brief evidence note is requested; the experiment does not collect private reasoning traces. Because the instructions form one treatment, their individual effects cannot be separated.

Direct elicitation requests every CPT row in one initial call. RNM requests only predetermined anchors and an aggregation family: weighted mean, weighted minimum, weighted maximum, or a minimum–maximum mixture. Unary targets use the identity family. The LLM supplies no weights or variance; these are fitted offline to its anchor distributions. Both methods receive the same influence signs, although only RNM imposes the corresponding monotonic aggregation constraint. Of the 92 retained parent relationships, nine signs were construct-supported, 43 based on qualitative context, and 40 explicitly marked as uncertain pre-collection hypotheses. Thus, the comparison includes the consequences of these frozen direction assumptions.

RAG consists of fixed, audited qualitative evidence cards. An offline retriever filters by source context, scores exact overlap between variable tags and the child-plus-parent set (wildcards contribute 0.25), and selects up to three positive-scoring cards, breaking ties by identifier. All 61 nodes receive one to three cards, totaling 24–103 text words (median 72) within a 220-word budget. Packets exclude reference probabilities, priors, weights, variances, numerical source results, and identifying links to reference artifacts. Measurement definitions remain permissible. Both arms already contain substantive definitions, so the treatment tests incremental qualitative information, not live retrieval or company-specific frequencies. Input-token budgets are not equalized.

3.3 Ranked-node reconstruction

Following the ranked-node construction [11, 15], each variable with c states partitions $[0, 1]$ into c equal intervals. Negative parent influences reverse semantic rank before aggregation; fitted probabilities are mapped back to the stored child order. Anchors include all-low, all-high, all-middle (rank $\lfloor (c-1)/2 \rfloor$ for each parent), each one-high/rest-low configuration, and each one-low/rest-high configuration, with duplicates removed. Selection uses no reference probabilities. For example, two ternary parents give nine rows, of which five are anchors: $(0, 0)$, $(2, 2)$, $(1, 1)$, $(2, 0)$, and $(0, 2)$. RNM reconstructs the other four; direct predictions are evaluated on those same four for RQ1.

For k parents, let G_r be the Cartesian product of five equally spaced, endpoint-inclusive points within each parent interval for row r. With standard-normal CDF Φ, child interval $[a_s, b_s]$, aggregation mean μ, and variance v, define

$$T_s(\mu, v) = \frac{\Phi((b_s - \mu)/\sqrt{v}) - \Phi((a_s - \mu)/\sqrt{v})}{\Phi((1 - \mu)/\sqrt{v}) - \Phi(-\mu/\sqrt{v})}. \tag{1}$$

The predicted probability is $q_{rs} = 5^{-k} \sum_{x \in G_r} T_s(f(x; w), v)$. Each truncated distribution is normalized before averaging. The mean functions are

$$f_{\text{mean}}(x; w) = \frac{\sum_i w_i x_i}{\sum_i w_i}, \tag{2}$$

$$g_i(x; w) = \frac{w_i x_i + \sum_{j \neq i} x_j}{w_i + k - 1}, \tag{3}$$

$$f_{\min} = \min_i g_i, \qquad f_{\max} = \max_i g_i, \tag{4}$$

$$f_{\text{mix}}(x; a) = (1 - a) \min_i x_i + a \max_i x_i. \tag{5}$$

The unary mean is x_1. Weighted-mean log weight ratios relative to the last parent lie in $[-6, 6]$; weighted-minimum/maximum weights lie in $[1, 20]$; $a \in [0, 1]$; and $v \in [10^{-4}, 1]$. Variance and minimum/maximum weights use logarithmic parameterizations.

For the family selected by the LLM, the fitter minimizes mean anchor total variation between fitted and elicited distributions. It uses eight deterministic starts (seed 42), coordinate-search steps 2^{-2} through 2^{-11} on normalized parameter coordinates, at most 20 sweeps per step, and an improvement threshold of 10^{-12}. Ties retain the earlier candidate. This seed controls fitting, not LLM sampling. The fitter accepts no reference CPT input. A poor valid fit remains an outcome; similar tables can arise from different parameters, so agreement does not establish parameter recovery. Reference-informed fits were computed separately after predictions were fixed, solely to diagnose approximation limits.

3.4 Execution, isolation, and validation

Five replications were collected on 15–16 October 2026 through signed-in Codex CLI 0.162.1 on a separate Windows computer, requesting `gpt-6-astra` with `medium` reasoning effort. The logs establish the requested configuration; actual serving-model identity was not independently verified. Each target–condition–replication used a fresh ephemeral session. No sampling temperature or generation seed was specified, and generation timeout and output limits remained client-managed. The original 616-template blocks were shuffled using Python's `Random(20261014+r)`, for replications $r = 1, \ldots, 5$. The retained 488 templates inherit those actual positions; they were not separately randomized after the scope restriction.

Reference files, full papers, evaluator code, and authoring conversations were excluded from the experimental computer/sessions. Browser, connector, memory, and tool controls were disabled and event logs inspected. Both arms received only their designated semantic inputs and evidence. This separation reduces direct access leakage, but cannot rule out memorization of public CPTs during pretraining or unobservable service context.

Validation checked identifiers, row completeness, state cardinalities, allowed evidence identifiers and function families, finite probabilities in $[0, 1]$, and row sums within 10^{-6} of one. Only residual normalization within tolerance was permitted and recorded. At most one fresh-session format repair supplied the original task,

its own answer, and validation errors, without reference feedback. The failure policy retained valid direct rows with uniform replacements for unresolved rows; unresolved RNM contracts or fitting failures received a uniform node fallback. Runtime/control anomalies paused collection for inspection. No fallback was required in the retained results.

3.5 Outcomes and statistical analysis

For reference row p and prediction q, the primary distance is

$$\mathrm{TV}(p, q) = \frac{1}{2} \sum_{s} |p_s - q_s| \in [0, 1], \tag{6}$$

where lower values indicate closer reference agreement. RQ1 uses identical non-anchor configurations for both methods. Within each replication, we average rows within targets, targets within BNs, and BNs within publication families, then average the four prompt and evidence combinations equally. Scrum is an equal-target mean reported separately; Chrome gives its two versions equal weight. We do not pool their primary effects.

RQ2 uses all rows and contrasts structured minus baseline or RAG minus no RAG within each method, averaging the other factor equally. Scrum remains separate; the external SE macro-average gives equal weight to four publication families. It is broader than the Chrome-only primary comparison. Secondary calculations retain Jensen–Shannon divergence in bits, mean absolute ordinal-CDF differences, and smoothed row-weighted CPT KL divergence. KL adds 10^{-8} to each state and renormalizes; its row weighting differs from the primary macro-average and it is not joint-distribution KL. The results below emphasize the frozen TV endpoint.

For RQ3, we count requested vectors and scalars before repairs, and aggregate client-reported input and output tokens across *all attempts*, including repairs. Total tokens are input plus output; cached-input and reasoning-output counters are subsets and are not added again. Paired effort contrasts use each method's replication totals, both for all 61 targets and the 25 primary targets. Summed request latency, repairs, and recorded offline fitting durations provide additional effort descriptions. Token counts from subscription execution are not converted into API-priced monetary savings.

The inferential unit is the replication block ($n = 5$), not a node, row, or treatment cell. We use two-sided paired Student t-tests with four degrees of freedom and pointwise 95% t intervals on block differences. These concern repeated executions on the fixed benchmark, assuming approximately normal, independent, stationary differences. Fresh sessions and shuffled order do not establish those assumptions, and five blocks provide little distributional evidence. A two-sided exact sign test checks sensitivity to block directions under independence; it addresses a direction/median null rather than exactly the same mean null. Even five concordant nonzero differences yield $p = 0.0625$.

Holm adjustments are applied separately to two primary contrasts, eight prompt/RAG contrasts, four token contrasts, and two same-anchor fitting contrasts. This is exploratory family-wise adjustment, not study-wide confirmatory control; intervals are not simultaneous. Other family and sensitivity comparisons are descriptive. No smallest effect of interest or equivalence margin was fixed, so significance does not establish practical usefulness and nonsignificance does not establish equivalence. No inference to arbitrary companies or unseen BNs is warranted.

3.6 Post-hoc offline diagnostics

With original predictions fixed, we compared uniform probabilities and secondary metrics, then varied reconstruction using saved responses and original anchor order. One variant fits direct judgments on paired RNM anchors with the RNM-selected family; others select among four families by LLM-anchor TV, increase search effort, or expand bounds. The supplement specifies search settings and bounds. Directions and five-point integration remain fixed; no reference probability enters fitting or selection.

Separately, reference-informed diagnostics minimize complete-reference TV across four families under stronger search, expanded bounds, and every combination of reversed parent directions. These are in-sample best-found approximations, not validated predictions or error floors. All additions are descriptive; original tests, predictions, collection tokens, and endpoints remain unchanged.

We also report Chrome version means and leave-one-node-out diagnostics, re-averaging remaining targets within BNs with equal Chrome-version weights.

4 Results

All five replications are complete: 488 retained initial trials per replication and 305 per treatment cell, totaling 2,440. Four format repairs (two per method) produce 2,444 attempts, with no uniform fallbacks. The audit rechecked 14,664 raw-file hashes and found no unexpected tool events or unresolved runtime errors. A service reconnect followed by a valid completed answer was retained without rerunning that request. Schema and control validation do not constitute an exhaustive human audit of evidence fidelity. All reported predictions and scores reuse the completed collection.

4.1 RQ1: Contrasting reference agreement

Table 3 reports the primary non-anchor comparisons separately. Table 4 shows all eight conditions, with the broader complete-table endpoints alongside them. Negative RNM-minus-direct differences favor RNM.

Scrum. RNM reduces primary TV from 0.3184 to 0.2761 (13.3% relative reduction; $t(4) = -18.68$). It is better in all five replications, on 12 of 14 targets, and in each matched prompt/evidence cell. This supports the hypothesized direction for the reconstructed Scrum reference under the paired-t assumptions (Table 3).

Chrome. RNM raises primary TV from 0.2223 to 0.2517 (13.2% relative increase; $t(4) = 23.37$), opposing the hypothesis. It is worse in all five replications and every matched cell, and better on only three of 11 targets. Neither stratum's exact sign test reaches 0.05. Thus, observed directions are stable, but rejection depends on small-sample parametric assumptions and does not establish practical importance. The supplement lists all block and target differences.

Chrome 40 TV changes from 0.2069 to 0.2102 ($\Delta = +0.0033$), versus 0.2377 to 0.2933 for Chrome 42 ($\Delta = +0.0556$). All 25 leave-one-node-out checks preserve their stratum's direction in all five blocks (supplement). Omitting Chrome 42's Major Contributors leaves Chrome at $\Delta = +0.0168$.

Table 3: Primary non-anchor TV, averaged across prompt/evidence conditions and five replications. Δ is RNM minus direct. Intervals are pointwise; Holm adjustment covers the two strata. Exact sign tests give raw $p = 0.0625$ and adjusted $p = 0.125$ for each.

Reference	Targets	Direct	RNM	Δ	95% CI for Δ	Raw t p	Holm t p
Scrum	14	0.3184	0.2761	−0.0423	[−0.0486, −0.0360]	4.83×10^{-5}	4.83×10^{-5}
Chrome	11	0.2223	0.2517	+0.0294	[+0.0259, +0.0329]	1.99×10^{-5}	3.97×10^{-5}

Table 4: Mean TV for all eight conditions. Primary columns use non-anchor rows of the 14 Scrum and 11 Chrome targets. Complete-table columns use all rows of 25 Scrum targets and 36 external SE targets; the latter averages four publication families, not only Chrome. No cross-stratum accuracy mean is formed.

Method	Prompt	Evidence	Scrum primary	Chrome primary	Scrum complete	External SE complete
Direct	Baseline	None	0.3180	0.2214	0.2620	0.1773
Direct	Baseline	RAG	0.3226	0.2194	0.2660	0.1813
Direct	Structured	None	0.3150	0.2259	0.2548	0.1751
Direct	Structured	RAG	0.3181	0.2225	0.2594	0.1743
RNM	Baseline	None	0.2792	0.2486	0.2353	0.2145
RNM	Baseline	RAG	0.2745	0.2539	0.2346	0.2166
RNM	Structured	None	0.2729	0.2509	0.2302	0.2152
RNM	Structured	RAG	0.2778	0.2534	0.2335	0.2147

Table 5: Diagnostics on identical anchor configurations of primary targets. Direct, raw RNM, and fitted RNM errors are against the reference. Fit residual compares fitted and raw RNM judgments. All use primary aggregation weights.

Quantity	Scrum	Chrome
Direct anchor TV	0.2914	0.2507
Raw RNM anchor TV	0.2790	0.2491
Fitted anchor TV	0.2762	0.2994
Fit residual TV	0.0678	0.1253
Reconstructed non-anchor TV	0.2761	0.2517

Table 6: Post-hoc primary non-anchor TV. Original outcomes are unchanged. Alternatives use saved judgments with no reference input to fitting or family selection.

Pipeline	Scrum	Chrome
Original direct	0.3184	0.2223
Original RNM	0.2761	0.2517
Direct anchors, paired RNM family	0.2924	0.2456
RNM anchors, family by fit	0.2787	0.2413
RNM, stronger search	0.2761	0.2517
RNM, expanded bounds	0.2761	0.2530

Elicited anchors versus fitted probabilities. Table 5 separates raw LLM judgments from RNM reconstruction on identical anchor configurations. On Scrum, fitting changes anchor TV only slightly, from 0.2790 to 0.2762 ($\Delta = -0.0028$, 95% CI $[-0.0053, -0.0003]$, raw and Holm t $p = 0.0352$). On Chrome it increases anchor TV from 0.2491 to 0.2994 ($\Delta = +0.0503$, CI $[0.0430, 0.0576]$, raw $p = 4.46 \times 10^{-5}$, Holm $p = 8.93 \times 10^{-5}$). Both adjusted sign-test values are 0.125. The Chrome deterioration therefore occurs even where the model was explicitly queried. Non-anchor and anchor errors concern different rows and cannot be subtracted as an additive error decomposition.

The direct-anchor comparison helps locate the Chrome disadvantage: raw RNM and direct judgments have similar anchor TV, whereas reconstruction worsens it. On Scrum, RNM anchors are already better than direct judgments before fitting. This is a pipeline comparison, not an isolated effect of requesting fewer rows.

A post-hoc uniform baseline gives primary TV of 0.4525 on Scrum and 0.3739 on Chrome, worse than both methods. The original contrast also persists descriptively under Jensen–Shannon divergence (Scrum: 0.1331 to 0.1066; Chrome: 0.0905 to 0.1104 bits) and ordinal-CDF distance (0.1887 to 0.1588; 0.1809 to 0.2000). These checks neither replace TV nor establish acceptable error for organizational decisions.

Offline sensitivity. Table 6 reports alternatives fitted only to saved LLM judgments. Using direct answers as anchors preserves the Scrum improvement and Chrome deterioration relative to original direct elicitation, while retaining the RNM-prompt-selected family. Family selection by anchor fit narrows Chrome's disadvantage but does not remove it. Stronger search and expanded bounds make little difference. These partial ablations preserve the contrast without isolating every prompt and fitting component.

Four Chrome targets have mean-direction reversals under frozen alignment and six have stochastic-order reversals. Scrum has two targets with reversals: a volatility-direction mismatch and a piecewise backlog relationship simplified by a global family. Excluding the former leaves the original Scrum contrast at −0.0402; Chrome's seven targets without mean reversals still give +0.0502. These descriptive checks do not justify changing the sample.

Reference-informed diagnostics fit complete reference tables directly. Stronger search over all four families gives best-found complete-table TV of 0.0560 for Scrum and 0.1592 for Chrome; expanded bounds give 0.0559 and 0.1589. Enumerating parent-direction flips further reduces these values to 0.0260 and 0.1322. Direction assumptions therefore contribute to mismatch, while a residual contrast remains in this search. These in-sample approximations use reference probabilities, are not blinded predictions or error floors, and do not certify global optima.

Illustrative configurations. We illustrate the all-middle anchor of the target nearest each stratum's median effect (full distributions in the supplement). For Scrum Team Orientation given Medium Personal Attributes and Expertise, reference, raw RNM, and fitted probabilities for Medium are 0.927, 0.585, and 0.586. For Chrome All Contributors given middle categories of Component Changes and Distinct Reviewers, they are 0.421, 0.396, and 0.537. Both select WMEAN in all 20 RNM trials. Middle-anchor TV improves despite opposite non-anchor effects; one configuration cannot explain an aggregate result.

For complete tables, Scrum TV changes from 0.2605 to 0.2334. The external family-specific changes are Chrome 0.2029 to 0.2278, APS 0.0964 to 0.1155, SESSI 0.1160 to 0.1143, and software logs 0.2927 to 0.4034. The equal-family external mean is 0.1770 versus 0.2153. Excluding the single SESSI target, which otherwise has one family weight, gives 0.1973 versus 0.2489 and preserves the direction. These descriptive endpoints cover all 61 targets, including unary and fully anchor-covered cases; they do not expand the primary compression sample.

4.2 RQ2: Structured prompts and qualitative RAG

Table 7 reports within-method effects on complete-table TV. For direct Scrum elicitation, structured prompting reduces TV by 0.0069, whereas qualitative RAG increases it by 0.0043; both survive the eight-test Holm correction under the paired-t model. The RNM Scrum contrasts do not survive correction. None of the four external SE macro-average contrasts survives correction, and no exact sign test in this eight-contrast family rejects at 0.05. The prompt and RAG hypotheses therefore receive no consistent support across methods and reference strata. These results concern the tested instruction bundle and qualitative packets; they do not establish equivalence or rule out benefits from informative numerical evidence.

4.3 RQ3: Accuracy and observed effort

Across all 61 targets, RNM requests 5,040 probability vectors instead of 8,420, a 40.1% reduction; requested scalars decrease from 24,740 to 14,700. This reduction is a property of the anchor design. Initial calls remain equal at 1,220 per method, as do attempts including repairs (1,222 each). Observed output-token reductions support the effort hypothesis, but their magnitude is smaller than the reduction in requested judgments (Table 8).

All-sample total-token savings do not survive testing, whereas both primary-target reductions survive the paired-t correction; no exact sign test rejects. Inputs dominate: 14,355,937 direct versus 14,307,719 RNM. Reasoning-output subsets are 9,890 versus 47,053. Shared inputs and RNM family-selection work are consistent with limited total-token savings, although these components were not independently manipulated.

Summed request latency is 14,696.6 seconds direct versus 14,135.2 RNM (3.82% lower), affected by service behavior, caching, and order. These are not separate method-only wall-clock times. Original reconstruction adds 182.3 seconds of recorded per-fit elapsed time over 1,059 distinct retained fit keys, not CPU time or SE-only wall time. Evidence preparation labor was not timed. The tradeoff is output compression with better Scrum and worse Chrome agreement.

5 Discussion

5.1 Research questions in relation to the literature

RQ1: Reference agreement. Scrum's advantage is consistent with useful reconstruction against an RNM-derived reference [11]; it persists with direct anchors. Chrome's raw anchor judgments are similarly accurate across methods, but fitting worsens agreement. The disadvantage concentrates in Chrome 42; no single target determines its direction. Numerical family selection narrows this disadvantage; stronger search and wider bounds do not remove it. This supports the distinction between ordinal eligibility and RNM compatibility [15]. Reference-informed direction changes reduce approximation error, but cannot separate remaining representation and optimization limitations. Chrome's documented interdependencies [14] provide context, not proof of the mechanism. Nor does CPT agreement establish defect-prediction accuracy.

RQ2: Elicitation support. Structured prompting benefits direct Scrum elicitation but not the other adjusted comparisons. Nafar et al. likewise distinguish query-granularity gains from prompt modifications [18]; their within-row queries and our across-row compression require matched evaluation. Qualitative packets give no consistent benefit beyond shared definitions. Clinical retrieval-usability findings [6] and reconstruction from statistical summaries [12] address different evidence and outcomes. Our results do not establish that informative quantitative retrieval is ineffective.

RQ3: Accuracy and effort. Fewer distributions yield lower output use, but small total-token savings. Extending the human burden–fidelity perspective [1], computational effort includes inputs, reasoning, repairs, and fitting. Compression improves reference agreement on Scrum while worsening it on Chrome, preventing a universal efficiency–accuracy recommendation.

5.2 Implications for SE researchers and practitioners

For researchers, the contrast motivates independently constructed references, validated influence directions, and multiple model configurations. Matched anchor and non-anchor comparisons distinguish elicitation from reconstruction effects. Hybrid inference such as BIRD [9] also motivates testing structured components rather than inferring fidelity from inspectability. Repeated runs do not establish generalization to other companies.

Table 7: RQ2: complete-table TV differences, averaging the other prompt/evidence factor equally. "Prompt" means structured minus baseline; "RAG" means RAG minus no RAG. External SE comprises four publication families. Negative differences favor the added support. Holm adjustment covers eight contrasts.

Reference stratum	Method	Contrast	Δ	Pointwise 95% CI	Raw t p	Holm t p
Scrum	Direct	Prompt	−0.00692	[−0.00843, −0.00541]	0.000220	0.001761
Scrum	Direct	RAG	+0.00430	[+0.00289, +0.00572]	0.001082	0.007575
Scrum	RNM	Prompt	−0.00313	[−0.00943, +0.00318]	0.240329	1.000000
Scrum	RNM	RAG	+0.00132	[−0.00237, +0.00500]	0.377615	1.000000
External SE	Direct	Prompt	−0.00462	[−0.01205, +0.00282]	0.159742	0.958452
External SE	Direct	RAG	+0.00163	[−0.00412, +0.00738]	0.475067	1.000000
External SE	RNM	Prompt	−0.00060	[−0.00798, +0.00678]	0.832498	1.000000
External SE	RNM	RAG	+0.00085	[−0.00556, +0.00726]	0.731511	1.000000

Table 8: RQ3: observed tokens including repairs. Totals span five replications; the difference and its interval refer to *one replication's* paired method totals. Workloads combine strata for accounting only, not for accuracy inference. Holm adjustment covers four contrasts. All exact sign tests give raw $p = 0.0625$, adjusted $p = 0.25$.

Targets	Counter	Direct total	RNM total	Reduction	Mean block Δ [95% CI]	Raw t p	Holm t p
All 61	Output	341,153	299,798	12.12%	−8,271 [−9,417, −7,125]	0.000037	0.000110
All 61	Input + output	14,697,090	14,607,517	0.61%	−17,915 [−39,331, +3,502]	0.080921	0.080921
Primary 25	Output	213,972	153,463	28.28%	−12,102 [−13,111, −11,093]	0.000005	0.000019
Primary 25	Input + output	6,384,519	6,196,992	2.94%	−37,505 [−58,833, −16,178]	0.008146	0.016292

For practitioners, RNM is a candidate initialization method requiring local validation. Review node meanings and influence assumptions, compare direct and reconstructed judgments, and validate additional configurations with company measurements or experts. Fit residuals measure agreement with judgments, not organizational outcomes. Tailoring reusable models remains plausible but requires held-out organizational validation before decision use.

6 Threats to Validity

Construct validity. TV measures contextual CPT agreement, not observed calibration, expert replacement, explanation quality, or decision utility. Equal-row scoring includes potentially rare configurations. Scrum's reconstruction lacks an independently certified AgenaRisk export, and its RNM origin favors the tested representation. Separate strata and anchor diagnostics expose this distinction. SESSI represents an original elicitation stage. Semantic adaptations and uncertain signs may affect agreement despite correct indexing.

Internal validity. Fresh sessions, shared definitions, hashes, and blinded fitting limit leakage. Public benchmark memorization remains possible; the alias does not establish the backend snapshot. Development involved Scrum. Offline variants cannot exhaust all explanations. Reference-informed selection is optimistic in-sample diagnosis; bounded search may miss better fits. Schema validation cannot establish semantic or evidence fidelity, and exhaustive human review was not performed. Packets and instructions remain treatment bundles.

Conclusion validity. Five blocks give little evidence about the distribution or stationarity of differences. The paired-t and exact-sign results must be considered together; neither treats shared nodes or rows as independent replications. Scope selection and testing were post hoc, and separate Holm families do not provide study-wide confirmatory error control. No prespecified acceptable-error margin supports claims of equivalence or practical sufficiency. Repeated execution estimates here are conditional on these fixed references and settings.

External validity. The primary comparison contains one external publication family (Chrome) and one Scrum model. The 61-target sample spans more settings, but many targets do not test multi-parent compression, and tables have at most 27 rows. Only one requested model and reasoning effort were evaluated. Organizational distributions, larger CPTs, other model families, and evidence containing local numerical measurements may yield different trade-offs. The excluded sensing and cyber-physical data do not support domain comparisons in this paper.

7 Conclusion

Across 61 ordinal SE targets, eight conditions, and five replications, RNM improved Scrum and worsened Chrome agreement on 25 primary targets. The original contrasts had paired-t support but no exact-sign rejection; saved-response variants preserved their directions. Prompting and qualitative evidence gave no consistent benefit; fewer judgments and output tokens yielded limited total-token savings.

Future work should test whether validated influence assumptions and organizational evidence make compact elicitation useful for local decisions.

Replication Materials

The package preserves original responses, predictions, tests, and usage, with separate post-hoc analyses and a supplement. Collection inputs exclude evaluator references. The repository is private for author review.

Use of Generative AI

LLM judgments are experimental data (Section 3). Generative AI assisted drafting and script preparation/checking; numerical claims trace to saved responses and deterministic analyses.

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